

Wi-Fi CSI Sensing — Application Landscape

Presence & Localization

Occupancy
detection

Indoor
localization

Person
counting

Activity Recognition

HAR
(walking, sitting...)

Gesture
recognition

Pose
estimation

Vital Signs Monitoring

Breathing
rate

Heart rate
& HRV

Sleep
monitoring

Safety & Automotive

Child presence
detection

Driver fatigue
monitoring

In-cabin activity
recognition

This thesis